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Automation Components and Devices

COMPARATIVE ANALYSIS OF ELECTRIC MOTORS AND DESIGN OF A RELAY-CONTACTOR CONTROL SYSTEM FOR THE MIXER DRIVE IN A WASHING POWDER PRODUCTION LINE

SIS №3

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INTRODUCTION

Automation of production processes is vital in the contemporary chemical industry to maintain product quality and ensure the safety of personnel. Among the most significant steps in the production chain of synthetic detergents, mainly laundry detergents, is the mixing process. The uniformity of the final product, its physical and chemical properties, as well as consumer attributes, are all directly dependent on the performance of the mixing unit (mixer). These units are powered by robust electric motors that are required to be reliable, energy-efficient, and resistant to difficult operating conditions.

This educational project concentrates on the comparison of industrial electric motors and the creation of a control system for the mixer drive in a laundry detergent production line. The topicality of this theme is due to the necessity of selecting power equipment that can endure tough operating conditions, which are usually accompanied by high starting torques, vibration, and possible dustiness. An incorrect motor choice may cause frequent emergency shutdowns, overheating of the equipment, and thus economic losses for the enterprise.

The goal of this project is to choose the best asynchronous electric motor of the two options (a Siemens and a Brook Crompton motor) and to come up with a control circuit design for the reliable relay-contactor operation. The project solves the tasks of: analyzing the motor datasheets, comparing the starting and operating characteristics, and confirming the choice of the most suitable model based on the specific load conditions of the mixer. The focus is also on the motor housing design and the extent to which it can muffle the mechanical vibrations.

The remaining part of the research is dedicated to the creation of a starting circuit. The large drive power (18.5 kW) makes it inadvisable to start the motor directly from the grid because of the high inrush currents, which cause voltage sags and new consumptions, thus can negatively impact the other equipment in the workshop. Therefore, the project implements a standard star-delta switching circuit that reduces the load on the grid during startup. The work also covers the calculation and the selection of the protective and switching components such as circuit breakers, contactors, and thermal relays along with picking the appropriate cables for the calculated current loads. The output from this work will be a comprehensive technical proposal that is production-ready and can be implemented without delay.

1. Selection of Reference Motors

To conduct a comparative analysis, two types of three-phase asynchronous motors were selected, which are shown in Table 1, which are technically capable of operating the mixer, but have fundamental differences in the rotor design.

Table 1 - Selected electric motors and a short description

Model	Type	URL
Siemens SIMOTICS SD (Severe Duty) 1LE1503-1EB2.	3-Phase Squirrel Cage Induction Motor (SCIM).	Link
BROOK CROMPTON BA4M025V4.	3-Phase Squirrel Cage Induction Motor (SCIM).	Link

2. Analysis of the characteristics of the selected motors

Asynchronous induction motors, which are the two models that we are comparing, basically work through the principle of electromagnetic induction. The motor stator has a three-phase winding, which is, by the way, the three phases are offset in space by 120 degrees each. After a three-phase voltage is connected, a rotating magnetic field (RMF) is created. The speed of this rotating field, N, is a function of the supply frequency, f, and the number of pole pairs, p, of the motor. For the two 4-pole machines (2 pole pairs) that we are comparing, at a normal supply frequency of 50 Hz in Kazakhstan, the synchronous speed can be found by the formula:

$$n_s = \frac{120 \cdot f}{P_{poles}} = \frac{120 \cdot 50}{4} = 1500 \text{ rpm}$$

This parameter is the theoretical speed limit the rotor approaches. However, for torque to be generated, relative motion between the stator field and the rotor conductors is required.

2.1 Siemens 1LE1503-1EB2 specifications analysis

The SIMOTICS SD (Severe Duty) 1LE1503 motor is the manufacturer's answer to the problem of heavy-duty applications. The 1LE1503 is a motor from the cast-iron series. One of the features of cast iron is that it absorbs vibrations well, which is very important for agitator drives, the shafts of which are subjected to considerable dynamic loads. Frame Size: 180 M. This is a standard IEC size that specifies the height of the rotation axis (180 mm). Here are all the main characteristics of this engine, as stated in its technical data sheet:

Table 2 – Characteristics of Siemens 1LE1503-1EB2

Parameter	Value	Comment
Nominal power (P_{nom})	18.5 kW	Powerful enough for a Mixer devise. It's perfect for this process.
Frame size (FS)	180 M	This is a standard IEC size that specifies the height of the rotation axis (180 mm)
Nominal voltage	400 V (Δ) / 690 V (Y)	This is a standard for motors of this power, allowing the use of Star-Delta starting.
Nominal torque (T_{nom})	120 Nm	$T_{nom} = \frac{P_{nom}}{\frac{n_{nom}}{60} \cdot 2\pi} = \frac{18500}{\frac{1470}{60} \cdot 2\pi} \approx 120\ Nm$, Which corresponds to the stated characteristics.
Nominal frequency (n_{nom})	1470 rpm	Normal for this type of motor.
Efficiency class	IE3 (92.6%)	(premium efficient) - has an even lower energy loss level than IE2 and is more energy efficient.
Power factor ($\cos(\phi)$)	0.82	This is a typical but not ideal value. Increased current consumption and wiring losses are possible.
Nominal current (I_{nom})	35 A (400 V (Δ))	$I_{nom} = \frac{P_{nom}}{\sqrt{3} \cdot V_{line} \cdot \eta \cdot \cos(\phi)} = \frac{18500}{\sqrt{3} \cdot 400 \cdot 0.926 \cdot 0.82} \approx 35\ A$, Which corresponds to the stated characteristics.
Starting current (I_{LR})	252 A (720%)	$I_{LR} = \left(\frac{I_{LR}}{I_{nom}}\right) \cdot I_{nom} = 720\% \cdot 35 = 252A$, This is a significant value, requiring special attention when selecting protective equipment (circuit breakers) and starting methods. Direct-on-line (DOL) starting will result in a 252 A surge current, which may cause a voltage drop in the network.
Starting torque (T_{LR})	300 Nm (250%)	$T_{LR} = \left(\frac{T_{LR}}{T_{nom}}\right) \cdot T_{nom} = 250\% \cdot 120 = 300\ Nm$, The engine has exceptional ability to overcome high static resistance. It is capable of starting very heavy mechanisms with a high moment of inertia.

Continuation of Table 2.

Parameter	Value	Comment
Break down torque (T_{BDT})	396 Nm (330%)	$T_{BDT} = \left(\frac{T_{BDT}}{T_{nom}}\right) \cdot T_{nom} = 330\% \cdot 120 = 396\ Nm,$ is exceptionally high and indicates a very high overload capacity.
Percent slip (s)	2%	$s = \frac{n_s - n_{nom}}{n_s} \times 100\% = \frac{1500-1470}{1500} \times 100\% = 2\%,$ This is a very low and excellent value for the operating mode of an asynchronous electric motor.
Service Factor (SF)	1.15	The device has an overload capacity of 15% above the rated power without failure
Duty Cycle (operating mode)	S1 (100%, Continuous)	Standard for mixer applications. Confirmed by SF 1.15. The duty type S1 can be defined as operation at a constant load maintained for sufficient time to allow the machine to reach thermal equilibrium.
Degree of protection	IP55	Protection against dust and water jets. Ideal for dusty laundry detergent production environments.
Thermal class	F (155 °C)	High heat resistance. Prevents overheating during long continuous mixing cycles under heavy load.
Mass and body material	165 kg (Cast Iron)	Heavy cast-iron construction absorbs vibrations effectively, providing essential stability for the dynamic mixer loads.
Noise level	66 dB(A)	Low noise level indicates precision balancing and high manufacturing quality, eliminating the need for soundproofing.

2.2 Brook Crompton BA4M025-4 specifications analysis

Brook Crompton is a famous British manufacturer. The BA series is the abbreviation for the Aluminum range. Housing material: Aluminum. Major difference from Siemens: Aluminium has better thermal conductivity than cast iron, but is less mechanically strong and less resistant to vibrations. Type: BA4M025-4. Energy: The "25" device is a 25 hp unit which is equal to 18.5 kW as per the specs. Here are all the main characteristics, as stated in its technical data sheet:

Table 3 – Characteristics of Brook Crompton BA4M025-4

Parameter	Value	Comment
Nominal power (P_{nom})	18.5 kW	Identical power to the reference motor. Sufficient for the mixer application.
Frame size (FS)	180 M	Standard IEC size, physically interchangeable with the Siemens motor.
Nominal voltage	400 V	Standard industrial voltage (3-phase).
Nominal torque (T_{nom})	120 Nm	$T_{nom} = \frac{P_{nom}}{\frac{n_{nom}}{60} \cdot 2\pi} = \frac{18500}{\frac{1470}{60} \cdot 2\pi} \approx 120\ Nm$, Standard torque for this power class.
Nominal frequency (n_{nom})	1470 rpm	Standard speed for a 4-pole motor at 50Hz.
Efficiency class	IE2 (91.2%)	Lower than Siemens (92.6% IE3). This indicates higher energy losses during operation.
Power factor ($\cos(\phi)$)	0.84	Slightly better than Siemens (0.82), meaning slightly more efficient use of incoming current.
Nominal current (I_{nom})	34.85 A (400 V)	$I_{nom} = \frac{P_{nom}}{\sqrt{3} \cdot V_{line} \cdot \eta \cdot \cos(\phi)} = \frac{18500}{\sqrt{3} \cdot 400 \cdot 0.912 \cdot 0.84} \approx 34.85\ A$, Calculated value based on power formula. The datasheet value (32.09 A) appears inconsistent with physics.
Starting current (I_{LR})	198.95 A (570%)	$I_{LR} = \left(\frac{I_{LR}}{I_{nom}}\right) \cdot I_{nom} = 570\% \cdot 34.85 = 198.95\ A$, This is significantly lower than the reference motor (7.2), providing a "softer" start for the grid.
Starting torque (T_{LR})	216 Nm (180%)	$T_{LR} = \left(\frac{T_{LR}}{T_{nom}}\right) \cdot T_{nom} = 180\% \cdot 120 = 216\ Nm$, This is lower than Siemens (300 Nm). Might be a disadvantage for starting a full mixer.
Break down torque (T_{BDT})	312 Nm (260%)	$T_{BDT} = \left(\frac{T_{BDT}}{T_{nom}}\right) \cdot T_{nom} = 260\% \cdot 120 = 312\ Nm$, Good overload capacity, but lower than Siemens (396 Nm).
Percent slip (s)	2%	$s = \frac{n_s - n_{nom}}{n_s} \times 100\% = \frac{1500-1470}{1500} \times 100\% = 2\%$, This is a very low and excellent value.

Continuation of Table 3.

Service Factor (SF)	1.0	The motor has no service factor reserve at 50Hz (unlike Siemens - 1.15). It should not be overloaded.
Duty Cycle (operating mode)	S1 (100%, Continuous)	Standard for continuous industrial processes like mixing.
Degree of protection	IP55	Protection against dust and water jets. Suitable for powder production environment.
Thermal class	F (155 °C)	High heat resistance. Prevents overheating during long continuous mixing cycles under heavy load.
Mass and body material	127 kg (Aluminium)	Lighter than Siemens (165 kg). Based on standard data for Brook Crompton Aluminum 180M frame. Low mass may lead to higher vibration in mixer applications compared to Cast Iron. Disadvantage for heavy duty. Unlike Siemens' Cast Iron, aluminum has lower vibration damping properties, which is critical for the mixer drive stability.
Noise level	70-76 dB(A)	Standard industrial value for this frame size is typically louder than the low-noise Siemens (66 dB).

3. Comparative Analysis and Selection of the Motor

Based on the calculations in Section 2, we can compare the Siemens 1LE1503-1EB2 with the Brook Crompton BA4M025-4 to figure out which one is the most appropriate to be used as a driver for a washing powder mixer.

3.1. Comparison by Key Criteria

<u>- Mechanical Stability (Vibration & Material):</u>

1. Siemens: The device is made with a heavy-duty cast iron frame, weighing 165kg. Besides, cast iron is known for its great vibration damping power.

2. Brook Crompton: The device comes with a lightweight aluminum frame, which weighs about 127 kg. Although aluminum is lighter, it is less resistant to the dynamic vibrations that arise from a heavy industrial mixer, thus, it is less effective in that aspect.

<u>- Starting Performance (Torque). The mixer is a high-inertia load that requires significant force to start, especially if the tank is full of powder mixture:</u>

1. Siemens: Delivers a short-term T_{LR} starting torque of 300 Nm (250%). This power is the one that definitely provides a strong start under load.

2. Brook Crompton: Delivers a starting torque of 216 Nm (180%). This is substantially lesser (by ~28%) and, therefore, entails the occurrence of a motor stall at startup if the mixture has high viscosity.

- Overload Capacity (Service Factor):

1. Siemens: Has a Service Factor (SF) of 1.15 meaning the device can be operated at 15% overload (up to ~21.2 kW) practically for unlimited periods without overheating. This is very important for sporadic variations in the powder density that may be realized.

2. Brook Crompton: Has an SF of 1.00, which means no safety margin for overload is provided.

- Energy Efficiency:

1. Siemens: Rated as IE3 (92.6%). For a continuous mode (S1), this leads to fewer power bills and less heat is generated.

2. Brook Crompton: Rated as IE2 (91.2%), which is less efficient.

- Grid Impact (Starting Current):

Here, Brook Crompton is better with a lower starting current ratio (5.7 vs. 7.2 for Siemens) that causes less impact or strain on the power grid. Nevertheless, this problem of power supply is not of a great magnitude for a single 18.5 kW motor, so it can be solved by simply using a Star-Delta starter, thus, making this advantage of Brook Crompton less critical than its torque one.

3.2. Final Selection

As a result of the comparative analysis, Siemens 1LE1503-1EB2 has been chosen as the best motor to be used in this project. Explanation of the washing powder production line: the manufacture of washing powder is a process of mixing viscous/granular materials, which results in uneven loads and vibrations. The Siemens motor is selected because:

- Cast Iron Body: It offers the required mechanical rigidity to absorb the mixer vibrations without breaking.

- High Starting Torque: The 300 Nm torque will definitely not let the mixer be stuck even if it is loaded to the maximum.

- Thermal Reserve: The Service Factor of 1.15 is like a "thermal" jacket for the motor, thus preventing it from overheating during peak loads, therefore, the production line is always reliable.

- Efficiency: The IE3 class is in line with present-day energy-saving standards.

Although the Brook Crompton motor features a "softer" start, its lower torque and absence of an overload reserve (SF 1.0) make it a risky choice for a heavy-duty application like this one.

4. Operating principle of the motor control circuit

The circuit that is outlined in the attached file allows for a "Star-Delta" start of an asynchronous motor, which is a typical practice for current inrush limitation. The story is as follows:

- Initial situation: Power is put on the input circuit breaker QF1. The control circuit is protected by fuse FU1. All contactors KM1, KM2, and KM3 are without tension, and the motor is at rest.
- Starting ("Start Mode"): Upon pressing the "START" button (SB2), the voltage is applied to the coil of the "Motor Power" contactor (KM1), "Start" contactor (KM3), and the time relay (KT1). Contact bors KM3 shut their power contacts, linking the motor windings in a star. The motor starts turning with a reduced voltage of 230 V being delivered to the winding; hence, the inrush current is limited to 84 A.
- Switching: Time relay KT1 is delaying the set time (accelerating time 5–8 seconds). The timer expires, the normally open contact KT1 breaks coil KM3 (star break). Upon the disconnection of KM3 by the contact, contact KT1 closes the circuit of contactor KM2. The "Delta" contactor (KM2) receives the power supply.
- Operating mode (Delta mode): The motor is connected to the full 400V mains voltage on the winding and thus, it is capable of delivering rated power and torque.
- Stop: Upon pressing the "STOP" button (SB1), the coil power circuit is disengaged; hence, all contactors are disconnected, and the motor is allowed to freewheel to a halt.

4.1 Protection Principle

- Short-circuit protection: It is achieved by a circuit breaker QF1 (with electromagnetic tripping) in the power circuit and a fuse FU1 in the control circuit.
- Overload protection: It is performed by a thermal tripping device in motor protection circuit breaker QF1 (or a separate thermal relay). When the current exceeds the rated value for a long time, the bimetallic strip heats up, and opens the auxiliary contact QF1.1, thus disconnecting the control circuit.
- Electrical interlock: The normally closed (NC) contacts KM2.1 and KM3.1 are used to prevent the star and delta contactors from being closed simultaneously thus, electrical short circuits are avoided.

4.2 Equipment specification

All equipment is listed below in Table 4.

Table 4 – Equipment specifications

№	Designation	Name	Specifications / Model
1	QF1	Automatic circuit breaker (3-pole)	Schneider Electric Acti9 iC60H 3P 50A, curve D (For starting motors)
2	KM1	Contactor (Main)	EKF KME 40A

Continuation of table 4.

№	Designation	Name	Specifications / Model
3	KM2	Contactor (Delta)	EKF KME 40A
4	KM3	Contactor (Star)	EKF KME 25A
5	KT1	Time relay (On delay)	Schneider Electric RE22
6	SB1	Control button (STOP)	IEK AEA-22
7	SB2	Control button (START)	IEK AEA-22
8	FU1	Fuse with holder	IEK ППНИ-33 00 6A IEK

CONCLUSION

This project required a comprehensive analysis of the requirements for a mixer electric drive in a washing powder production line and the creation of a complete control system for this mechanism. The project was broken down into several logical stages, each of which made it possible to substantiate the technical solutions adopted.

As the first stage, an in-depth comparison was made between two industrial asynchronous motors of the same rated power: the Siemens 1LE1503 series and the Brook Crompton BA series. The comparison has revealed that, while the electrical parameters of the motors are almost identical, the designs of the two motors significantly differ, and these differences are crucial for operation under vibration loads. The Siemens motor, in a cast iron housing, was found to have a number of advantages over its aluminum Brook Crompton counterpart due to its higher weight and better damping properties. In addition, the Siemens motor chosen has a higher IE3 energy efficiency class and, what is more, a higher starting torque, thus ensuring the loaded mixer is started in a safe and reliable manner. The decision has been made to go with the Siemens motor based on a combination of factors (reliability, starting characteristics, and overload margin).

The control circuit was the second major element of the project. Because direct starting of a motor of such power results in a sevenfold current surge, a circuit with star-delta winding switching was devised. This approach cut the starting current to about one-third, thus allowing the motor to be started smoothly and the power supply to be protected from overloads. The circuit employs relay-contactor logic which is simple and reliable, without the use of costly frequency converters, thus making the project economically viable and easy to service. To ensure safety, the contactors are electrically interlocked, so that a short circuit cannot occur when switching modes.

The last stage of the project was about figuring out the currents and picking the real-life hardware. They chose modern protection and switching pieces that are up to international standards. The cable cross-section was chosen with a safety margin, not only considering the prolonged heating under the rated load but also short-term overloads during startup. The control system developed fully satisfies the technical specifications, provides the required level of protection for the equipment and personnel (including short-circuit protection and motor thermal protection), and is the guarantee for the stable operation of the washing powder production line in continuous operation.

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